Fire Risk Prediction by Building and by Month
Final Report

Mireille EL Haddad

Problem Statement

Each year, fire incidents in Montréal cause significant damage, loss, and fear. With over 300,000 addresses in the city, the fire department faces a monumental challenge: how to allocate limited inspection resources to prevent the most fires, in the right places, at the right times.

Traditionally, fire inspections follow fixed schedules or rely on intuition and past experience. But fires are not random—they follow patterns, shaped by a building's structure, its neighborhood, and its history. We asked ourselves: *What if we could anticipate where fires are likely to happen next month—before they occur?*

This project is the result of that question. Our goal was to develop a predictive tool to support the Montréal Fire Department in proactively identifying high-risk buildings, month by month, using data they already have and insights they may not yet be leveraging.

Data Sources

This project is based on open data from the City of Montréal and combines property, fire incident, and geospatial address information to support monthly fire risk prediction. The *Unités d'évaluation foncière* dataset provides structural building details. Fire response records are from the *Interventions du Service de sécurité incendie de Montréal* (SIM), and geolocation is enabled via the *Adresses ponctuelles* dataset. Together, they enable spatiotemporal modeling of fire risk per building, per month.

Additional data sources, such as crime data from the City of Montreal and weather data, were explored to enrich the prediction model. Preliminary analysis suggested that crime density and certain weather patterns might correlate with fire occurrence. Population census data was also identified as a potentially valuable source for understanding socioeconomic context at the neighborhood level. However, due to time constraints and the additional complexity of integrating these temporal and spatial features, they were not included in the final feature set. These datasets remain promising for future iterations of the project.

Data Exploration and Cleaning

To develop a monthly fire risk prediction model for Montréal, a comprehensive cleaning and merging process was applied to property, fire intervention, and address datasets. The goal was to produce a spatiotemporal, building-level panel dataset.

Property Assessment Data Cleaning
The original property dataset contained 512,288 unique address records. The cleaning script (dataprep/evaluation_fonciere.py) removed entries with invalid construction years (before 1800 or after 2025) and imputed missing values using borough- and usage-based medians. Specifically, NOMBRE_LOGEMENT had ~13.5% missing values, ETAGE_HORS_SOL ~14.8%,

and SUPERFICIE_BATIMENT ~17.2%. Column names and data types were standardized for consistent processing. The cleaned output was saved as eval_cleaned_feat_eng.csv

Fire Intervention Records Cleaning

The intervention dataset was built by concatenating two official SIM datasets covering the periods 2020–2022 and 2023 to the date (check date of downloading). Using the script interventions_HAS_FIRE_binary_analysis.py, the CREATION_DATE_TIME field was cleaned and standardized to ISO format. Only confirmed fire-related incidents were retained by filtering records with "INCENDIE" or "AUTREFEU" in the DESCRIPTION_GROUPE field, while unrelated categories such as "Alarmes-incendies" were excluded to avoid inflating fire counts. This filtering step yielded a total of 21,421 confirmed fire incidents, with 14,650 incidents related to AUTREFEU and 6,771 related to INCENDIE . Columns with internal coordinates (MTM8_X, MTM8_Y) were dropped. The final cleaned dataset was saved as interventions_cleaned_with_has_fire.csv, and will serve as the geospatial fire event source for subsequent modeling steps.

Spatial Integration and Dataset Merging

Using dataprep/main_evaluation_feat_eng.py, the cleaned property, address, and fire datasets were merged. Geolocation was performed by matching civic numbers and cleaned street names, successfully assigning coordinates to 461,882 buildings (Rows with valid coordinates). The remaining 201,901 without spatial data were flagged using a missing_coords indicator.

A 100-meter spatial buffer was applied around each fire incident, and buildings within those buffers were linked to fire events, resulting in 294,767 buildings marked as having experienced at least one fire. Time features such as fire_month, fire_season, and year_month were derived from the fire date.

At the borough level, 2024 fire statistics were computed for Montréal's 19 zones, including FIRE_COUNT_LAST_YEAR_ZONE and FIRE_RATE_ZONE, which were normalized using MinMaxScaler to support machine learning modeling. Additional features related to building structure and density were also included.

The final dataset, saved as evaluation_fire_coordinates_date_feat_eng_2.csv, contains 663,783 rows. While each row represents a building, some may be duplicated due to multiple fire records. This dataset is designed as a foundation for building a monthly panel dataset to forecast fire risk at the building level.

Feature Engineering

To enable monthly fire risk prediction, features were engineered to capture each building's structural, spatial, historical, and temporal characteristics.

Static attributes include:

- MUNICIPALITE
- ETAGE_HORS_SOL
- NOMBRE_LOGEMENT
- AGE_BATIMENT

as well as land use codes like

- CODE_UTILISATION
- CATEGORIE_UEF.

Surface measures such as

- SUPERFICIE_TERRAIN
- SUPERFICIE_BATIMENT

are complemented by derived metrics including

- RATIO_SURFACE
- DENSITE_LOGEMENT
- HAS_MULTIPLE_LOGEMENTS

Zone-level fire activity is summarized through

- FIRE_COUNT_LAST_YEAR_ZONE,
- FIRE_RATE_ZONE, FIRE_FREQUENCY_ZONE

their normalized versions (e.g., FIRE_RATE_ZONE_NORM), and BUILDING_COUNT.

Recent fire history is represented by rolling or cumulative metrics like

- fire_rolling_3m, fire_rolling_6m
- fire_rolling_12m
- fire_cumcount.

Temporal context is introduced via **month_num** and **year**, enabling the model to capture seasonality and long-term patterns.

Together, these features provide a multi-dimensional representation of fire risk across Montréal's building stock.

Tools and Techniques Used

This project was developed entirely in Python, using open-source tools well-suited for handling large, structured, and geospatial datasets.

- **pandas** and **NumPy** were used for data wrangling and transformation.
- **geopandas** and **shapely** enabled spatial joins and buffering to associate fires with nearby buildings within 100 meters.
- **scikit-learn** provided tools for feature scaling, model evaluation, and baseline classification models.
- **LightGBM** and **XGBoost** were used to train high-performance models capable of handling imbalanced data.

Fire risk predictions were visualized with interactive **Folium heatmaps**.
The full project code and documentation are available on GitHub:
🔗 https://github.com/mireillehaddad/Fire-Risk-Montreal-Xgboost

Summary of Modelling Techniques Evaluated

We began our modeling work by testing Random Forests and LightGBM on two types of data structures: a flat format and a panel format. In the flat setup—where each row represented a building—modeling fire events across multiple months proved problematic. A single building might be linked to multiple fire months, resulting in ambiguous and hard-to-interpret labels.

Transitioning to a **panel structure**, where each row corresponds to a specific building and month, resolved this issue. It allowed us to treat fire prediction as a clear **binary classification task**: *Did a fire occur in this building during this month?*

All models were trained on historical data from **January 2020 to December 2023** and tested on **2024**—a setup chosen to mimic real-world deployment and avoid **temporal data leakage**, which could otherwise inflate performance metrics.

Because only about **1%** of building-months involve a fire, the task posed a **severe class imbalance**. Our primary focus was **recall**—capturing as many actual fire cases as possible. At the same time, we tracked **precision** to avoid overwhelming inspectors with false positives. After tuning and comparison, **XGBoost** emerged as the best-performing model, using `scale_pos_weight` to compensate for imbalance. While it achieved high recall, it struggled with precision—especially when using fixed probability thresholds.

To make the system practical, we shifted to **Precision@k**, which ranks buildings by predicted risk and focuses inspection efforts on the top-k highest-risk cases. This approach aligns with the city's limited inspection capacity and ensures that the model is used as a decision support tool rather than a rigid classifier.

Modelling Results

Panel data structure - XGBoost

XGBoost was applied to the monthly building-level panel to predict fire occurrence (HAS_FIRE_THIS_MONTH). The target variable is binary, indicating whether a fire occurred in a given building during a specific month.

Features Used

Multiple iterations of the model were trained with various subsets of these features. In the end, using fire_rolling_3m, fire_rolling_6m, fire_rolling_12m leads to little improvement and makes the model less usable in the future (since recent history of the future is not known yet), so it is best to ignore fire_rolling_3m, 6m and 12m.

Model Configuration

The model was trained using XGBClassifier with:

- scale_pos_weight=124 (to address class imbalance)
- n_estimators=100, max_depth=5, learning_rate=0.1
- enable_categorical=True, eval_metric="logloss", random_state=42

Training was done on data from 2020–2023; testing was on 2024.

Performance Summary

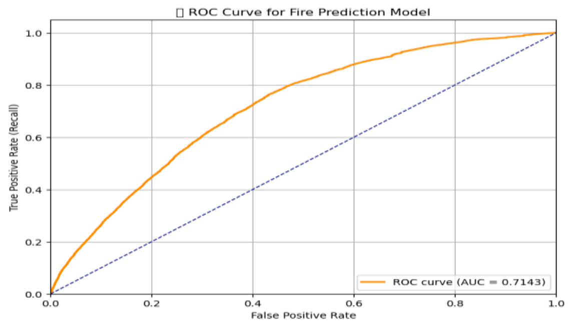
Threshold	Recall	Precision	F2 Score	TP	FP	FN
0.50	60.4%	2.7%	0.051	~19,000	~700,000	~31,000
0.20	95.6%	1.7%	0.065	~30,339	~1.1M	~1,300

At the default threshold (0.5), the model captures 60.4% of fires but suffers from low precision (2.7%). Lowering the threshold to 0.2 dramatically increases recall (95.6%) at the cost of increased false positives, reducing precision to 1.7%.

Interpretation

XGBoost is effective at identifying fire cases, especially at lower thresholds. However, its precision remains low, making it more suitable for **risk ranking** rather than binary classification. A practical use case would be to prioritize inspections based on predicted probabilities (e.g., top k% of buildings), rather than relying on fixed cutoffs.

This ROC curve supports the idea that the model is better suited for ranking buildings by risk, not hard classification.



➢ The model achieved an AUC of 0.714, indicating moderate predictive power. This means there's a 71.4% chance it ranks a random fire case higher than a non-fire case.

➢ The ROC curve rises well above the diagonal, especially at low thresholds—showing the model captures many true positives with relatively few false alarms.

➢ Implication:
The model is useful for ranking fire risk and prioritizing inspections, but in highly imbalanced settings like this, precision-recall curves offer a more informative evaluation.

Feature importance

We did not apply dimensionality reduction techniques like PCA, so all input features remain interpretable. While some features are correlated—such as nombre_logement and has_multiple_logements, or the various rolling fire history metrics—they do not dominate the top-ranked features. Thus, feature importance remains meaningful.

Approximately 50% of the model's predictive power comes from just five features:

- ratio_logement (building-to-lot size ratio)
- nombre_logement (number of units)
- no_arrond_ILE_CUM (borough)
- fire_cumcount (cumulative local fire count)
- densite_logement (unit density per building size)

	Feature	Gini Importance
9	RATIO_SURFACE	0.345957
2	NOMBRE_LOGEMENT	0.132729
8	NO_ARROND_ILE_CUM	0.093800
16	fire_cumcount	0.050082
10	DENSITE_LOGEMENT	0.049593
3	AGE_BATIMENT	0.041110
19	fire_rolling_12m	0.037208
5	CATEGORIE_UEF	0.034448
17	fire_rolling_3m	0.032469
21	year	0.031738
20	month_num	0.025620
1	ETAGE_HORS_SOL	0.024897
0	MUNICIPALITE	0.024021
6	SUPERFICIE_TERRAIN	0.022491
18	fire_rolling_6m	0.021934
4	CODE_UTILISATION	0.017497
7	SUPERFICIE_BATIMENT	0.014405
11	HAS_MULTIPLE_LOGEMENTS	0.000000
13	BUILDING_COUNT	0.000000
12	FIRE_FREQUENCY_ZONE	0.000000
14	FIRE_RATE_ZONE_NORM	0.000000
15	FIRE_COUNT_LAST_YEAR_ZONE_NORM	0.000000

Surprisingly, building usage type (residential, commercial, etc.) played only a minor role in prediction.

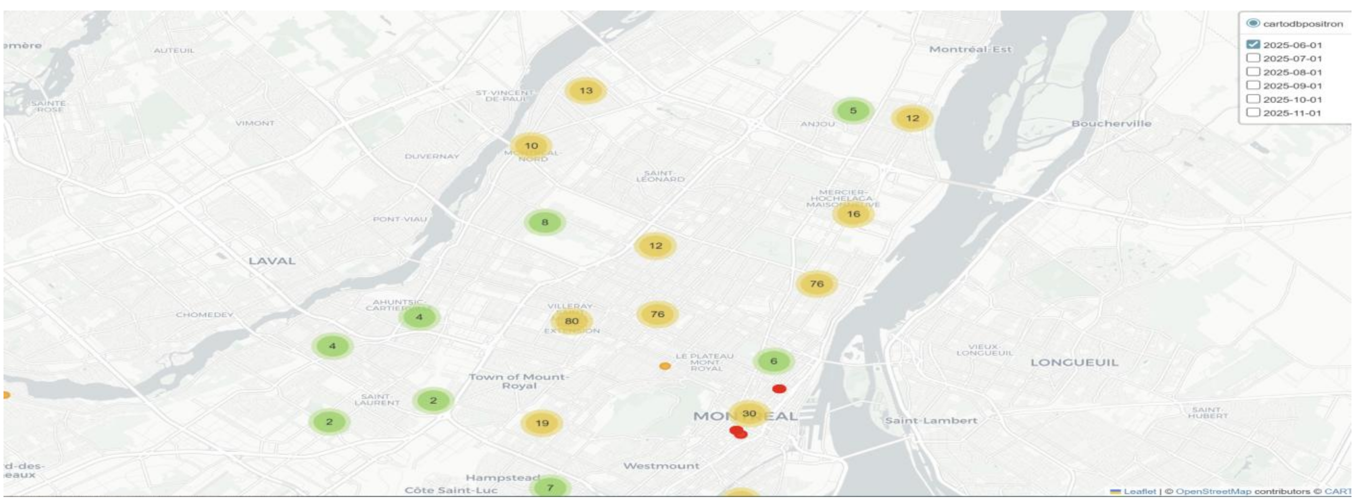
Prioritization of high risk buildings

The following table shows a comparison of performing 7500 random inspections vs prioritizing inspections based on the highest predicted probability for the full year. The "Actual Fire" column indicates an actual fire within a 100m radius which is why even the random pick finds a high number of "actual fires". Using our model to predict fire risk drastically improves the efficiency of fire inspections (47.84% of buildings inspected would have had a fire in 100m radius in 2024 vs 14% with random pick)

Method	Actual fire*	Precision@K
Random pick	1058	14.11%
Highest predicted risk	3588	47.84%

Note: 7500 is the inspection capacity stated by the fire department in the project introductory video.

Visualizations



Heat maps and business context [Montreal Fire Risk Maps]

Multiple fire risk maps were developed throughout the project, but the final interactive version stands out as the most informative. It serves as a practical tool for prioritizing building inspections based on predicted monthly fire risk.

Business Context from the Fire Risk Map

This interactive map is designed not just to display risk—but to support informed, data-driven decision-making for fire prevention. Buildings are color-coded by risk level: **red dots** indicate the **highest-risk buildings**, while **orange dots** represent **moderate-risk cases**. This intuitive design helps quickly identify where inspections should be prioritized.

Targeted Inspection Planning: The map displays the **top 500 highest-risk buildings** for any selected month. Through the **dropdown menu**, users can adjust both the **month** and the **number of buildings to prioritize (k)**—allowing fire prevention teams to develop **flexible, monthly inspection plans** tailored to resource availability. For example, smaller teams may focus on the top 200 buildings, while full-capacity teams might target the top 1,000. Boroughs with persistently high risk can be prioritized for deeper inspections.

Monthly Risk Monitoring and Planning : The map's **month-by-month filtering** enables teams to track how fire risk shifts throughout the year. Certain zones may consistently appear in the top-k risk range during specific months. These patterns support **rolling monthly inspection strategies**, helping allocate resources in advance and proactively address seasonal or temporal trends.

Stakeholder Communication & Reporting : The map also serves as a transparent, accessible tool for communicating risk and strategy. It helps inspection planners present their decisions to city officials, justify resource requests, and demonstrate the rationale for focusing on specific neighborhoods. This builds trust in the model and improves stakeholder alignment.

Scenario Analysis and Dynamic Adjustment : By allowing users to toggle the number of top-risk buildings displayed, the map supports **scenario-based planning**. Decision-makers can explore how inspection outcomes might change depending on available staffing or shifts in risk levels—making it easier to **balance depth of inspection with breadth of coverage**.

From Prediction to Monthly Action : With its combination of predictive risk scores and interactive filters, the map becomes a practical tool for building and adjusting monthly inspection plans. It turns raw model outputs into meaningful, location-based strategies—helping Montréal prevent fires more efficiently and equitably.

Insights and Challenges

1. **Panel Data Was Key to Temporal Risk Modeling**
   The shift to a monthly panel structure—where each row represents a building-month pair—proved essential. This approach allowed the model to reflect changes in fire risk

over time and made the classification task more meaningful and interpretable than a flat dataset.

2. **Risk Ranking (Precision@k) Was More Useful Than Thresholds**
 Given the operational need to prioritize limited inspections, Precision@k outperformed traditional metrics like accuracy or even fixed-threshold recall. By ranking buildings by predicted risk and inspecting only the top-k, the model delivered a major efficiency gain: 47.8% of top-ranked buildings had fires within a 100m radius in 2024, compared to just 14% using random selection.

3. **Data Imbalance Was the Most Significant Modeling Challenge**
 Fires occurred in only about 1% of all building-month records, creating a highly imbalanced dataset. This imbalance made it difficult for standard classifiers to detect rare fire events—they defaulted to predicting "no fire," yielding high accuracy but low usefulness.
 We addressed this by tuning scale_pos_weight in XGBoost and focusing evaluation on **recall and precision**, not accuracy. Still, balancing both metrics remained difficult: increasing recall often led to lower precision and vice versa. This trade-off is central to deploying an effective and actionable model.

4. **Spatiotemporal Integration Required Careful Engineering**
 Aligning building, address, and fire incident data on a monthly basis while preserving temporal integrity was complex. Building correct join logic, validating fire-month associations, and managing duplicates added significant engineering overhead.

5. **Fire Location Data Was Approximate**
 Because fire incidents were not tied to building-level coordinates, we used 100-meter spatial buffers to associate fires with buildings. While this enabled risk labeling, it may have introduced false positives by assigning fire risk to nearby but unrelated structures.

6. **External Data Could Not Be Integrated Within Scope**
 Although we prepared weather, crime, and population datasets, time limitations prevented full integration. These data sources remain promising for future model iterations.

Despite these challenges, the project successfully delivered a functioning prototype that supports real-world inspection planning, offering both predictive power and actionable insights for the City of Montréal.

Conclusions

This project demonstrated the feasibility and value of predicting monthly fire risk at the building level across Montréal. By integrating property, fire incident, and address data, and applying a time-aware panel approach with machine learning, the team was able to produce actionable risk scores and visualize them on an interactive map.

The model, while not perfectly precise, is highly effective at capturing true fire events—making it well-suited for inspection prioritization. The ability to adjust predictions by month and focus on the top-k riskiest buildings empowers the fire department to proactively plan inspections based on current and evolving risk. Importantly, this project sets the stage for future iterations. With more precise fire location data and integration of external variables like weather, crime, and socioeconomic indicators, the predictive power and operational value of the model could be significantly enhanced. Ultimately, this work supports a shift from reactive fire prevention toward proactive, data-driven public safety planning.

Future Work

To enhance both the accuracy and real-world utility of the model, several next steps are recommended:

1. Refine Fire Attribution:
The current method links fires to buildings using a 100-meter buffer, which may inflate positive labels. Improved geocoding or access to address-level incident data would enhance precision.

2. Integrate External Risk Factors:
Incorporating crime (e.g., arson, vandalism) and weather variables (e.g., wind, temperature) could strengthen predictions by capturing environmental and social drivers of fire risk.

3. Grade Risk Using Broader Event Types:
Future models could assign varying risk levels based on related categories like "AUTREFEU" and frequent alarms, shifting from binary prediction to graded risk scoring.

4. Improve Interpretability and Trust:
Using tools like SHAP or LIME would help explain why specific buildings are flagged, supporting transparency and stakeholder confidence.

5. Enable Dynamic Planning and Real-Time Adaptation:
Incorporating lagged features or evolving to sequential models (e.g., LSTM, transformers) could improve responsiveness to recent fire trends. Real-time data integration would allow for a timely inspection response.

6. Build a Full Operational Dashboard:
Beyond the interactive map, a dedicated dashboard could let inspection teams filter buildings by borough and month, annotate cases, export lists, and monitor trends—bridging the gap between predictions and daily decision-making.

7. Incorporate Weather Dataset:
We used a time series model to forecast fire incidents and examine patterns over time. The

results showed a clear downward trend and repeating seasonal peaks. While we prepared weather data to potentially include in the analysis, we did not use it in this model because of limited time. Future work could look more closely at how weather affects fire risk.

References

[Unités d'évaluation foncière - Site web des données ouvertes de la Ville de Montréal]

https://donnees.montreal.ca/en/dataset/interventions-service-securite-incendie-montreal

https://donnees.montreal.ca/en/dataset/adresses-ponctuelles?activity_id=bf1841a6-2965-497b-8599-b9465e6472c8

https://donnees.montreal.ca/dataset/casernes-pompiers

https://donnees.montreal.ca/dataset/requete-311

Appendix

Other models performance

Two types of modeling approaches were explored during this project: models trained on a flat data structure and models trained on a panel dataset with monthly building-level granularity.

For the **flat data structure**, a baseline was established using:

- **Random Forest**, with imbalance correction applied through the class_weight parameter
- **LGBM**

Flat data structure LGBMClassifier

Features used

ETAGE_HORS_SOL	ANNEE_CONSTRUCTION	SUPERFICIE_BATIMENT
NOMBRE_LOGEMENT	SUPERFICIE_TERRAIN	LONGITUDE
		LATITUDE

Results

	Precision	**Recall**	**F1-Score**	**Support**
Accuracy			0.434	58954
Macro Avg	0.439	0.433	0.434	58954
Weighted Avg	0.437	0.434	0.433	58954

Flat data structure - Random forest classifier

Features used

Log_terrain	log_etage_hors_sol	AGE_BATIMENT
log_batiment	log_numbre_de_logement	density

Results

	precision	recall	f1-score	support
1	0.208	0.153	0.176	5328
2	0.204	0.198	0.201	4670
3	0.234	0.181	0.204	5406
4	0.256	0.191	0.219	5881
5	0.227	0.162	0.189	6266
6	0.219	0.196	0.207	5071
7	0.258	0.303	0.279	4450
8	0.183	0.216	0.198	4384
9	0.202	0.248	0.222	4056
10	0.211	0.213	0.212	4575
11	0.173	0.186	0.180	4558
12	0.217	0.223	0.220	4499
13	0.874	0.917	0.895	73613
accuracy			0.599	132757
macro avg	0.267	0.261	0.262	132757
weighted avg	0.582	0.599	0.589	132757

	Precision	Recall	F1-Score	Support
Accuracy			0.599	132757
Macro Avg	0.267	0.261	0.262	132757
Weighted Avg	0.582	0.599	0.589	132757

The flat data structure poses a multilabel classification problem in the sense that a given x vector can be part of multiple fire_month classes (for instance, there may have been a fire in february and may so the same building x in X can be classified as both class 2 and 5. Training on this multilabel dataset as if it was a single label multiclass classification led to poor results. To overcome this, the target variable should be transformed or we could explore the use of the n_outputs parameter of sklearn RandomForestClassifier.

Panel data structure: it allows for time-aware prediction at the building-month level, and was used to develop more robust models. The following techniques were tested:

- **Random Forest**, with oversampling strategies including **SMOTE** (Synthetic Minority Oversampling Technique) and BalancedRandomForestClassifier from imblearn,
- **XGBoostClassifier**, with threshold tuning for imbalanced classification
- **LGBMClassifier**

Random Forest with panel data :

Simple Random Forest (no compensation for imbalanced dataset) : Recall 9.30%

	precision	recall	f1-score	support
0	0.988	0.931	0.958	3674405
1	0.027	0.139	0.045	50239
accuracy			0.920	3724644
macro avg	0.507	0.535	0.502	3724644
weighted avg	0.975	0.920	0.946	3724644

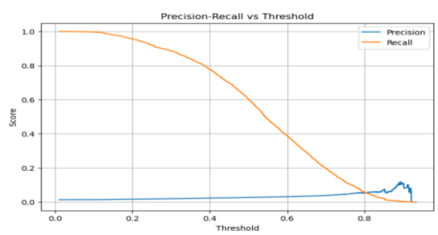
Using imblearn.BalancedRandomForestClassifier :

SMOTE oversampling gave an even worse result than the simple RandomForest. (result was not kept)

Even the best attempt (BalancedRandomForest) is nowhere near the results obtained with XGBoost so we continued with XGBoost.

To support model evaluation and operational decision-making, We did the following visualizations that we did not include in the report :

- **Precision–Recall Curve**

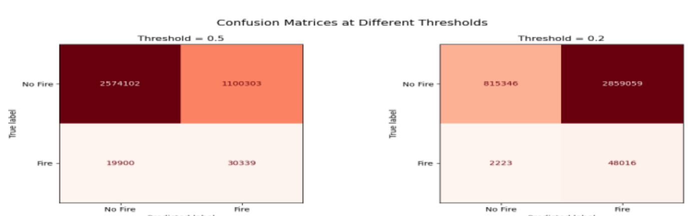


This plot shows how precision and recall vary as you change the prediction threshold.

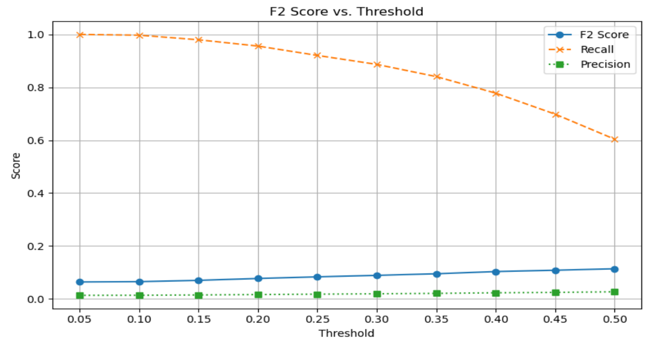
- **At low thresholds (e.g., 0.2), the model detects most fires (high recall) but with many false alarms (low precision).**

- **At higher thresholds (e.g., 0.5+), it becomes more precise but misses many fires. Because precision stays low overall, the model is better for ranking buildings by risk than for strict yes/no fire prediction.**

- **Confusion Matrices** (at 0.2 and 0.5 thresholds)



- **F2 Score vs. Threshold plot**

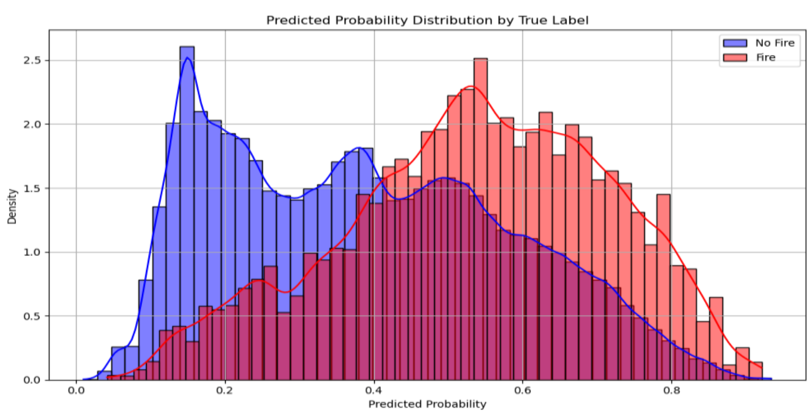


- ➢ **Recall (orange): Starts high (~100%) at low thresholds and steadily decreases. This is expected—lower thresholds classify more fires correctly, but also more false positives.**

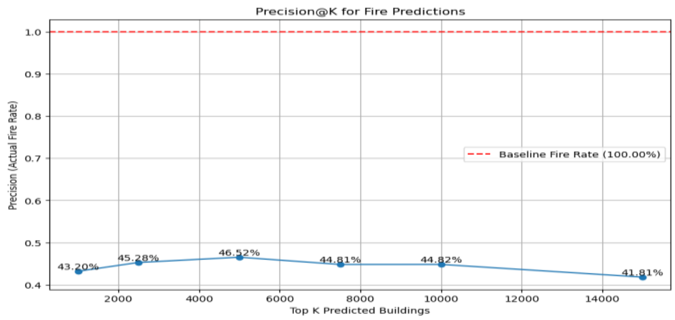
- ➢ **Precision (green): Very low and slowly increases with threshold. This means many false positives exist, especially at low thresholds.**

- ➢ **F2 Score (blue): Peaks around thresholds 0.45–0.50. Since F2 gives more weight to recall, it favors thresholds that still capture many fires but with slightly improved precision.**

- **Predicted probability distribution**



This plot displays the predicted probability distributions for buildings with and without fires. Buildings without fires tend to have lower predicted probabilities (around 0.1–0.4), while buildings with fires show higher probabilities (centered around 0.5–0.8). Although there is some overlap, the separation indicates that the model assigns higher risk scores to true fire cases, supporting its use for ranking buildings by fire risk rather than making strict binary classifications.



This chart shows how precision (i.e., actual fire occurrence rate) varies across the top-K most fire-prone buildings as ranked by the model. Precision peaks around the top 5,000 predictions (46.52%) and then gradually declines, indicating that the model is most confident in its top predictions. The red dashed line represents the baseline fire rate across all buildings, showing that the model performs significantly better than random selection.